MTH2008 – Real Analysis

Jan 2026 – Solutions

SECTION A

1. (a) (i) State the definition of the upper Riemann sum of a bounded function $f : [0,1] \to \mathbb{R}$ with respect to a partition $P = (x_0, x_1, \ldots, x_n)$ of $[0,1]$. (5)

Solution: The upper Riemann sum of f with respect to P is

$$U(f;P) = \sum_{i=1}^{n} \sup\{f(x) \mid x_{i-1} \leq x \leq x_i\}\,(x_i - x_{i-1}),$$

(ii) State Riemann's criterion for integrability of a bounded function $f : [0,1] \to \mathbb{R}$. (5)

Solution: The function f is Riemann integrable if and only if, it is bounded, and for all $\epsilon > 0$, there exists a partition P of $[0,1]$ such that

$$U(f;P) - L(f;P) < \epsilon.$$

(iii) For $n \in \mathbb{N}$, let $f_n : [0,1] \to \mathbb{R}$. State sufficient conditions under which

$$\lim_{n\to\infty} \int_0^1 f_n(x)\,\mathrm{d}x = \int_0^1 \lim_{n\to\infty} f_n(x)\,\mathrm{d}x. \quad (5)$$

Solution: If for each $n \in \mathbb{N}$, the function f_n is Riemann integrable, and if the sequence of functions $(f_n)_{n\in\mathbb{N}}$ converges uniformly to a well-defined function f, then

$$\lim_{n\to\infty} \int_0^1 f_n(x)\,\mathrm{d}x = \int_0^1 \lim_{n\to\infty} f_n(x)\,\mathrm{d}x.$$

(iv) Give an example of a bounded non-Riemann integrable function. (5)

Solution: An example of a non-Riemann integrable function is the function $f : [0,1] \to \mathbb{R}$ defined, for $x \in [0,1]$, by

$$f(x) = \begin{cases} 1 & \text{if } x \text{ is rational,} \\ 0 & \text{if } x \text{ is irrational.} \end{cases}$$

(v) Give a justification for your answer to Part (iv). (5)

Solution: To see that the function f given in the solution to Part (iv) is non-Riemann integrable, let $P = (x_0, x_1, \ldots, x_n)$ be a partition of $[0, 1]$. Since the rationals are dense in $[0, 1]$, we have that

$$\begin{aligned} L(f; P) &= \sum_{i=1}^{n} \inf\{f(x) \mid x_{i-1} \leq x \leq x_i\} (x_i - x_{i-1}) \\ &= \sum_{i=1}^{n} 0 \cdot (x_i - x_{i-1}) = 0. \end{aligned}$$

Moreover, since the irrationals are dense in $[0, 1]$, we have that

$$\begin{aligned} U(f; P) &= \sum_{i=1}^{n} \sup\{f(x) \mid x_{i-1} \leq x \leq x_i\} (x_i - x_{i-1}) \\ &= \sum_{i=1}^{n} 1 \cdot (x_i - x_{i-1}) = \sum_{i=1}^{n} (x_i - x_{i-1}) = 1. \end{aligned}$$

Since this holds for any partition, it follows that

$$\underline{\int_0^1} f(x)\,\mathrm{d}x = \sup_P L(f; P) = 0 \neq 1 = \inf_P U(f; P) = \overline{\int_0^1} f(x)\,\mathrm{d}x,$$

and therefore, f is not Riemann integrable.

(b) (i) State the mean value theorem. (5)

Solution: Let $a, b \in \mathbb{R}$ with $a < b$. If f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b), then there exists $c \in (a, b)$ with

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

(ii) Define what it means for a function $f : [0, 1] \to \mathbb{R}$ to be Lipschitz continuous. (5)

Solution: The function f is Lipschitz continuous if and only if there exists $M \in \mathbb{R}$ such that, for all $x, y \in [0, 1]$,

$$|f(x) - f(y)| \leq M|x - y|.$$

(iii) Let the function $f : [-1, 1] \to \mathbb{R}$ be defined, for $x \in [-1, 1]$, by

$$f(x) = |x|.$$

Prove that f is Lipschitz continuous but not differentiable. (5)

Solution: For $x, y \in [-1, 1]$, by the definition of f and the reverse triangle inequality,

$$|f(x) - f(y)| = ||x| - |y|| \leq |x - y|.$$

Thus, f is Lipschitz continuous with Lipschitz constant 1. To see that f is not differentiable, we show f is not differentiable at 0. To this end observe, given $\epsilon > 0$, $x \in (0, \epsilon)$ and $y \in (-\epsilon, 0)$, that

$$\frac{f(x) - f(0)}{x - 0} = \frac{x}{x} = 1 \quad \text{and} \quad \frac{f(y) - f(0)}{y - 0} = \frac{|y|}{y} = -1.$$

Thus, the limit

$$\lim_{z \to 0} \frac{f(z) - f(0)}{z - 0}$$

does not exist and so f is not differentiable at 0.

(iv) Let $f : [0,1] \to \mathbb{R}$ be a continuous function which is differentiable on $(0,1)$ with $f(0) = f(1) = 0$. State the conclusion of Rolle's theorem for the function f. (5)

Solution: There exists $c \in (0,1)$ with $f'(c) = 0$.

(v) Let $g : [0,1] \to \mathbb{R}$ denote a continuous function that is differentiable on $(0,1)$. Further, suppose that there exists $M \in \mathbb{R}$ with

$$|g'(x)| \leq M$$

for all $x \in (0,1)$. Use the mean value theorem to show that g is Lipschitz continuous. Justify your answer. (5)

Solution: By the hypothesis of the question, the function f satisfies the hypotheses of the mean value theorem, and so, given $x, y \in [0,1]$ with $x < y$, there exists $c \in (x,y)$ with

$$|g(y) - g(x)| = |g'(c) \cdot (x - y)| = |g'(c)| \cdot |x - y|.$$

This in tandem with the condition that there exists $M \in \mathbb{R}$ with

$$|g'(x)| \leq M$$

for all $x \in (0,1)$, yields that

$$|g(y) - g(x)| \leq M \cdot |x - y|.$$

SECTION B

2. (a) State what it means for a function $g : [0,1] \to \mathbb{R}$ to be differentiable at a point $x \in (0,1)$, and what it means for g to be differentiable on $(0,1)$. (5)

Solution: The function g is said to be differentiable at a point $x \in (0,1)$ if the following limit exists

$$\lim_{y \to x} \frac{g(x) - g(y)}{x - y}.$$

The function g is said to be differentiable on $(0,1)$ if it is differentiable at every point in $(0,1)$.

(b) Let $f : \mathbb{R} \to \mathbb{R}$ be defined, for $x \in \mathbb{R}$, by

$$f(x) = \begin{cases} x^2 \sin(1/x) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

(i) Prove that f is continuous at zero. (5)

Solution: Given $\epsilon > 0$, for all non-zero $x \in \mathbb{R}$ with $|x| < \epsilon^{1/2}$,

$$|x^2 \sin(1/x) - 0| = |x|^2 \cdot |\sin(1/x)| \leq |x|^2 < \epsilon,$$

yielding the required result.

(ii) Prove that f is differentiable everywhere on its domain, and compute its derivative. (5)

Solution: Since compositions and products of differentiable functions are differentiable, and since polynomials, rational functions (provided the denominator is non-zero) and trigonometric functions are differentiable, it follows that for all non-zero $x \in \mathbb{R}$ the function f is differentiable at x with derivative

$$\frac{\mathrm{d}}{\mathrm{d}x} f(x) = 2\,x\,\sin(1/x) - \cos(1/x).$$

Here, in computing the derivative, we have used the product rule and chain rule for differentiation.
To compute the derivative of f at zero, we calculate the following limit:

$$\lim_{x \to 0} \frac{x^2 \sin(1/x) - 0}{x - 0} = \lim_{x \to 0} x \sin(1/x).$$

Observe that, given $\epsilon > 0$, for all non-zero $x \in \mathbb{R}$ with $|x| < \epsilon$,

$$|x \sin(1/x) - 0| = |x| \cdot |\sin(1/x)| \leq |x| < \epsilon.$$

With this observation at hand, we obtain that $f'(0) = 0$.

(iii) Determine if the derivative of f is differentiable. Justify your answer. (10)

Solution: Since compositions and products of differentiable functions are differentiable, and since polynomials, rational functions (provided the denominator is non-zero) and trigonometric functions are differentiable, it follows that for all non-zero $x \in \mathbb{R}$ the derivative of f is differentiable at x. Therefore, to determine if the derivative of f is differentiable, we need only check if the derivative of f is differentiable at zero. To this end, we consider the limit

$$\lim_{x\to 0} \frac{(2x\sin(1/x) - \cos(1/x)) - 0}{x - 0} = \lim_{x\to 0} \left(2\sin(1/x) - \frac{\cos(1/x)}{x}\right).$$

Since for $n \in \mathbb{N}$,

$$2\sin(1/(2/(\pi + 4\pi n))) - \frac{\cos(1/(2/(\pi + 4\pi n)))}{2/(\pi + 4\pi n)} = 2$$

and

$$2\sin(1/(2/(3\pi + 4\pi n))) - \frac{\cos(1/(2/(3\pi + 4\pi n)))}{2/(3\pi + 4\pi n)} = -2$$

it follows that

$$\lim_{x\to 0} \frac{(2x\sin(1/x) - \cos(1/x)) - 0}{x - 0} = \lim_{x\to 0} \left(2\sin(1/x) - \frac{\cos(1/x)}{x}\right)$$

does not exist, and thus the derivative of f is not differentiable at zero, and hence not differentiable.

3. (a) State what it means for a sequence of real numbers $(a_n)_{n\in\mathbb{N}}$ to be Cauchy. (5)

Solution: A sequence of real numbers $(a_n)_{n\in\mathbb{N}}$ is Cauchy, if given a real number $\epsilon > 0$, there exists a natural number N, such that $|a_n - a_m| \leq \epsilon$ for all natural numbers $n, m \geq N$.

(b) Let $(a_n)_{n\in\mathbb{N}}$ denote a sequence of real numbers.

(i) Prove, using Cauchy's convergence criterion for series, that if the series $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim_{n\to\infty} a_n = 0$. (5)

(ii) Prove, or give a counterexample to the statement, if $\lim_{n\to\infty} a_n = 0$, then $\sum_{n=1}^{\infty} a_n$ is a convergent series. If giving a counterexample, you may give it without justification. (5)

Solution:

(i) Given $\epsilon \in (0, \infty)$, since $\sum_{n=1}^{\infty} a_n$ converges, it follows from Cauchy's convergence criterion for series that there exists $N \in \mathbb{N}$ such that for all integers $m, n \geq N$ with $m \geq n$,

$$|a_n + a_{n+1} + ... + a_m| < \epsilon.$$

In particular, given an integer $n \geq N$, and letting $m = n$,

$$|a_n - 0| = |a_n| < \epsilon.$$

However, this is precisely what it means for the sequence $(a_n)_{n\in\mathbb{N}}$ to converge to 0.

(ii) The statement is false. Consider the sequence $(a_n)_{n\in\mathbb{N}}$ with $a_n = 1/n$. In this case, we have that

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} 1/n = 0,$$

but that the series

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} 1/n$$

is divergent, namely is not convergent.

(c) Using Cauchy's convergence criterion for series, determine for which values of $p \in \mathbb{R}$, with $p > 0$, the following series is convergent.

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{k^p}$$

Justify your answer. (10)

Solution: Fix $p \in \mathbb{R}$ with $p > 0$. For n an even natural number, observe that

$$\frac{(-1)^n}{n^p} + \frac{(-1)^{n+1}}{(n+1)^p} > 0 \quad \text{and} \quad \frac{(-1)^{n+1}}{(n+1)^p} + \frac{(-1)^{n+2}}{(n+2)^p} < 0.$$

Thus, if n is an even natural number, for all natural numbers $m \geq n$, we have

$$\underbrace{\frac{(-1)^n}{n^p} + \frac{(-1)^{n+1}}{(n+1)^p}}_{\geq 0} + \cdots + \frac{(-1)^m}{m^p} > \begin{cases} \dfrac{1}{m^p} & \text{if } m \text{ is even,} \\ 0 & \text{if } m \text{ is odd,} \end{cases}$$

and

$$\frac{(-1)^n}{n^p} + \underbrace{\frac{(-1)^{n+1}}{(n+1)^p} + \frac{(-1)^{n+2}}{(n+2)^p}}_{<0} + \cdots + \frac{(-1)^m}{m^p} < \begin{cases} \dfrac{1}{n^p} & \text{if } m \text{ is even,} \\ \dfrac{1}{n^p} + \dfrac{1}{m^p} & \text{if } m \text{ is odd.} \end{cases}$$

Thus, we may conclude, for $n, m \in \mathbb{N}$, with $n \leq m$ and n even, that

$$\left| \frac{(-1)^n}{n^p} + \frac{(-1)^{n+1}}{(n+1)^p} + \frac{(-1)^{n+2}}{(n+2)^p} + \cdots + \frac{(-1)^m}{m^p} \right| \leq \frac{2}{n^p}.$$

This in tandem with the triangle inequality yields, for $n, m \in \mathbb{N}$, with $n \leq m$ and n odd, that

$$\left| \frac{(-1)^n}{n^p} + \frac{(-1)^{n+1}}{(n+1)^p} + \frac{(-1)^{n+2}}{(n+2)^p} + \cdots + \frac{(-1)^m}{m^p} \right| \leq \frac{3}{n^p}.$$

Hence, given an $\epsilon > 0$, letting $N > (3/\epsilon)^{1/p}$, for all natural numbers $n, m > N$ with $m \geq n$,

$$\left| \frac{(-1)^n}{n^p} + \frac{(-1)^{n+1}}{(n+1)^p} + \frac{(-1)^{n+2}}{(n+2)^p} + \cdots + \frac{(-1)^m}{m^p} \right| \leq \frac{3}{n^p} \leq \frac{3}{N^p} < \epsilon.$$

Namely that Cauchy's convergence criterion for series is satisfied, and so the series converges. Thus, the given series converges for all $p \in \mathbb{R}$ with $p > 0$.

4. (a) State what it means for a function $f : [0,1] \to \mathbb{R}$ to be monotonically increasing. (5)

Solution: The function f is said to be monotonically increasing if $f(x) \leq f(y)$ for all $x, y \in [0,1]$ with $x \leq y$.

(b) Prove that if $f : [0,1] \to \mathbb{R}$ is a monotonically increasing function, then f is Riemann integrable. (5)

Solution: Let $n \in \mathbb{N}$, and let $P_n = (x_0 = 0, x_1 = 1/n, x_2 = 2/n, \ldots, x_n = 1)$ be a partition of $[0,1]$, and recall that the norm of the partition P_n is given by

$$\|P_n\| = \sup\{x_i - x_{i-1} \mid i \in \{1, 2, \ldots, n\}\} = 1/n.$$

Further, since f is monotonically increasing on $[0,1]$, we have, for all $x, y \in [0,1]$ with $x < y$, that $f(y) \geq f(x)$. Therefore,

$$\begin{aligned} L(f; P_n) &= \sum_{i=1}^{n} \inf\{f(x) \mid x_{i-1} \leq x \leq x_i\} (x_i - x_{i-1}) \\ &= \sum_{i=1}^{n} f(x_{i-1}) \|P_n\| = \frac{1}{n} \sum_{i=1}^{n} f(x_{i-1}) \end{aligned}$$

and

$$\begin{aligned} U(f; P_n) &= \sum_{i=1}^{n} \sup\{f(x) \mid x_{i-1} \leq x \leq x_i\} (x_i - x_{i-1}) \\ &= \sum_{i=1}^{n} f(x_i) \|P_n\| = \frac{1}{n} \sum_{i=1}^{n} f(x_i). \end{aligned}$$

Hence,

$$\begin{aligned} U(f; P_n) - L(f; P_n) &= \left(\frac{1}{n} \sum_{i=1}^{n} f(x_i) \right) - \left(\frac{1}{n} \sum_{i=1}^{n} f(x_{i-1}) \right) \\ &= \frac{1}{n} \sum_{i=1}^{n} (f(x_i) - f(x_{i-1})) = \frac{1}{n} (f(1) - f(0)). \end{aligned}$$

Thus, for a given $\epsilon > 0$, if $n \in \mathbb{N}$ is such that $n > (f(1) - f(0))/\epsilon$, we have that

$$U(f; P_n) - L(f; P_n) = \frac{1}{n} (f(1) - f(0)) < \epsilon.$$

Hence, Riemann's criterion for integrability is satisfied, meaning that f is Riemann integrable.

(c) For $k \in \mathbb{N}$, define $\chi_k : [0,1] \to \mathbb{R}$ by

$$\chi_k(x) = \begin{cases} 0 & \text{if } x \in [0, 2^{-k}), \\ 1 & \text{if } x \in [2^{-k}, 1]. \end{cases}$$

For fixed $n \in \mathbb{N}$, determine if the function $g_n : [0,1] \to \mathbb{R}$ defined, for $x \in [0,1]$, by

$$g_n(x) = \sum_{k=1}^{n} 2^{-k} \chi_k(x)$$

is Riemann integrable. Justify your answer. (5)

Solution: For $k \in \mathbb{N}$, observe that χ_k is monotonically increasing. Indeed, if $x, y \in [0, 2^{-n})$ or if $x, y \in [2^{-n}, 1]$, then $\chi_k(y) = \chi_k(x)$, and if $x \in [0, 2^{-n})$ and $y \in [2^{-n}, 1]$, then $\chi_k(y) = 1 > 0 = \chi_k(x)$. Thus, by Part (a), for $k \in \mathbb{N}$, the function χ_k is Riemann integrable. Since a finite sum of Riemann integrable functions is again Riemann integrable, we have that

$$g_n(x) = \sum_{k=1}^{n} 2^{-k} \chi_k(x)$$

is Riemann integrable, as is it is a finite sum of Riemann integrable functions.

(d) For $k \in \mathbb{N}$, let $\chi_k : [0,1] \to \mathbb{R}$ be as defined in Part (c).
Determine if the function $g : [0,1] \to \mathbb{R}$ defined, for $x \in [0,1]$, by

$$g(x) = \sum_{k=1}^{\infty} 2^{-k}\, \chi_k(x)$$

is well defined and Riemann integrable. Justify your answer. (10)

Solution: To see that g is well defined, observe that $g(0) = 0$,

$$g(1) = \sum_{k=1}^{\infty} 2^{-k} = 1,$$

and that if $x \in (0,1)$, then there exists an $n \in \mathbb{N}$ with $x \in [2^{-n}, 2^{-n+1})$ and so

$$g(x) = \sum_{k=n}^{\infty} 2^{-k} = \frac{1}{2^{n-1}},$$

yielding that g is well-defined. To see that g is Riemann integrable, recall that, if a sequence of functions $(f_n)_{n\in\mathbb{N}}$, with common domain $[0,1]$, converge uniformly, then

$$\lim_{n\to\infty} \int_0^1 f_n(x)\,\mathrm{d}x = \int_0^1 \lim_{n\to\infty} f_n(x)\,\mathrm{d}x.$$

Namely, the limiting function is Riemann integrable. Thus, to conclude that g is Riemann integrable, it suffices to show that the sequence of functions $(g_n)_{n\in\mathbb{N}}$ converges uniformly to g. Observe that

$$\begin{aligned} \|g_n - g\|_\infty &= \left\| \sum_{k=1}^{n} 2^{-k}\,\chi_k - \sum_{k=1}^{\infty} 2^{-k}\,\chi_k \right\|_\infty \\ &= \left\| \sum_{k=n+1}^{\infty} 2^{-k}\,\chi_k \right\|_\infty = \sum_{k=n+1}^{\infty} 2^{-k} = \frac{1}{2^n}. \end{aligned}$$

Therefore, given an $\epsilon > 0$, letting $N > -\ln(\epsilon)/\ln(2)$, for all integers $n > N$,

$$\|g_n - g\|_\infty = \frac{1}{2^n} < \epsilon.$$

In other words, the sequence of functions $(g_n)_{n\in\mathbb{N}}$ converges uniformly to g, and thus g is Riemann integrable.